

Assignment 1: Shell Programming

Operating System Lab (Unix)

Problem Statement

Write shell scripts to perform basic operations in a Unix environment. The programs should demonstrate understanding of shell scripting concepts including variables, conditions, control structures, and functions.

Tasks to Implement

1. Create and execute a basic shell script that prints a message
2. Write a script to accept two numbers and display their sum
3. Write a script to check whether a number is even or odd
4. Write a script to find the largest of three numbers
5. Write a script to display numbers from 1 to N using a loop
6. Write a script to calculate factorial of a number
7. Write a script using functions to perform basic arithmetic operations

Instructions

- Create script files using `.sh` extension
- Make the script executable using `chmod +x filename.sh`
- Execute scripts using `./filename.sh`
- Use variables, conditional statements (`if`), and loops (`for`, `while`)
- Use functions where required
- Provide meaningful output messages

Sample Input / Output

Example Run:

```
Enter first number: 5
Enter second number: 3

Sum = 8
```

Example (Even/Odd):

```
Enter number: 4

4 is Even
```

Constraints

- Use only shell scripting (no external programming languages)
- Ensure scripts run without errors
- Validate user input where necessary
- Follow proper syntax and indentation

Bonus Tasks (Optional but Recommended)

1. Write a script to check whether a number is prime
2. Write a script to display Fibonacci series
3. Write a script to count number of files in a directory
4. Write a script to display system information (date, user, working directory)
5. Write a menu-driven shell script